

Eric Culf

Personal Information

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Education

PhD in Applied Mathematics - Quantum Information

University of Waterloo

January 2023 – August 2026

Master of Science in Mathematics and Statistics with Concentration in Mathematics

University of Ottawa

January 2021 – December 2022

Honours Bachelor of Science with Major in Physics and Major in Mathematics

University of Ottawa

September 2016 – December 2020

Selected Awards and Scholarships

- NSERC A. G. Bell Canada Graduate Scholarship — Doctoral 2024 - 2026
- IQC Achievement Award 2025
- University of Waterloo President's Graduate Scholarship 2025
- Xanadu Graduate Prize in Quantum Photonics and Information 2023
- uOttawa Mathematics and Statistics Outstanding Student Paper Award 2022 - 2023
- NSERC Canada Graduate Scholarship — Master's 2021 - 2022
- D.K.C. MacDonald Memorial Prize 2020
- Faculty Plaque in Physics 2020
- Armen Manoogian Memorial Scholarship 2017 – 2018

Work Experience

University of Waterloo Teaching Assistant

- September 2025 - December 2025: Teaching assistant/corrector for the course Introduction to Quantum Information Processing (QIC 710)
- September 2024 - December 2024: Teaching assistant/corrector for the course Introduction to Quantum Information Processing (QIC 710)

- September 2023 - December 2023: Teaching assistant/corrector for the course Introduction to Quantum Information Processing (QIC 710)
- January 2023 - April 2023: Teaching assistant/corrector for the course Real Analysis with Applications (AMATH331)

University of Ottawa Teaching Assistant

- September 2022 - December 2022: Teaching assistant for the course Raisonnement mathématique et preuves (MAT1762); Teaching assistant for the course Calcul différentiel et intégral II (MAT1722)
- January 2022 - April 2022: Teaching assistant/corrector for the course Raisonnement mathématique et preuves (MAT1762); Teaching assistant for the course Introduction à l'algèbre linéaire (MAT1741)
- September 2021 - December 2021: Teaching assistant/corrector for the course Introduction to linear algebra (MAT1341); Teaching assistant for the course Calcul différentiel et intégral 1 (MAT1720)
- January 2021 - April 2021: Teaching assistant/corrector for the course Raisonnement mathématique et preuves (MAT1762)

NSERC Undergraduate Student Research Award

- May 2020 - August 2020: Summer student at the University of Ottawa Department of Mathematics under the supervision of Professor Anne Broadbent
- May 2019 – August 2019: Summer student at the University of Ottawa Department of Mathematics under the supervision of Professor Abdellah Sebbar

National Research Council Student Employment Program

- May 2018 – August 2018: Summer student with the Ultrafast Quantum Optics group (Security and Disruptive Technologies)
- May 2017 – August 2017: Summer student at the Data Analytics Centre (Information and Communication Technology)

Academic Service

- Program committee member for QCrypt 2025
- Subreviewer for FOCS 2025
- Subreviewer for LatinCrypt 2025
- Subreviewer for QIP 2025
- Reviewer for Journal of the ACM, 2024
- Subreviewer for QCrypt 2024
- Subreviewer for QIP 2024

Publications

- E. Culf, J. van Dobben de Bruyn, M. Vernooij, P. Zeman (2025). “Existence and nonexistence of commutativity gadgets for entangled CSPs.” ArXiv Preprint.
- A. Bhattacharyya, E. Culf (2025). “Uncloneable Encryption from Decoupling.” Accepted to Nature Physics.
- M. Brannan, E. Culf, M. Vernooij (2024). “Quantum expanders and property (T) discrete quantum groups.” ArXiv Preprint.
- P. Botteron, A. Broadbent, E. Culf, I. Nechita, C. Pellegrini, D. Rochette (2024). “Towards Unconditional Uncloneable Encryption.” ArXiv Preprint.
- E. Culf, K. Mastel (2024). “RE-completeness of entangled constraint satisfaction problems.” Accepted to Proceedings of the 2025 IEEE 66th Annual Symposium on Foundations of Computer Science.
- V. Asadi, R. Cleve, E. Culf, A. May (2024). “Linear gate bounds against natural functions for position-verification.” Quantum 9, 1604.
- V. Asadi, E. Culf, A. May (2024). “Rank lower bounds on non-local quantum computation.” Proceedings of the 16th Innovations in Theoretical Computer Science Conference. LIPIcs 325, 11.
- E. Culf, H. Mousavi, T. Spirig (2023). “Approximation algorithms for noncommutative CSPs.” Proceedings of the 2024 IEEE 65th Annual Symposium on Foundations of Computer Science.
- E. Culf, A. Mehta (2023). “New Approaches to Complexity via Quantum Graphs.” Accepted to Quantum Information and Computation.
- A. Broadbent, E. Culf (2023). “Uncloneable Cryptographic Primitives with Interaction.” ArXiv Preprint.
- A. Broadbent, E. Culf (2023). “Rigidity for Monogamy-of-Entanglement Games.” Proceedings of the 14th Innovations in Theoretical Computer Science Conference. LIPIcs 251, 28.
- E. Culf, T. Vidick, V. Albert (2022). “Group coset monogamy games and an application to device-independent continuous-variable QKD.” ArXiv Preprint.
- E. Culf, T. Vidick (2022). “A monogamy-of-entanglement game for subspace coset states.” Quantum 6, 791.
- Q. Exirifard, E. Culf, E. Karimi (2021). “Towards communication in a curved space-time geometry.” Communications Physics 4, 171.
- J. Rico, A. Nantel, L. Phuong Pham, R. Voyer, Y. Durocher, S. Penny, A. Surendra, D. Pinto, M. Sasseville, E. Culf, S. Leclerc, M. VanHetHoog, M. Cuperlovic-Culf (2017). “In silico modelling and metabolomics for optimization of monoclonal antibody fed-batch production in CHO cells.” BMC Systems Biology.

Presentations

- [E. Culf](#), [K. Mastel](#). “RE-completeness of entangled constraint satisfaction problems.” Talk at TQC 2025, September 2025, Bengaluru, India.
- M. Brannan, [E. Culf](#), M. Vernooij. “Quantum Expanders and property (T) quantum groups.” Talk at OAQI 2025, June 2025, Waterloo, Canada.
- [E. Culf](#), [H. Mousavi](#), [T. Spirig](#). “Approximation algorithms for noncommutative CSPs.” Talk at QIP 2025, February 2025, Raleigh, North Carolina, USA.
- [E. Culf](#), [K. Mastel](#). “RE-completeness of entangled constraint satisfaction problems.” Poster at QIP 2025, February 2025, Raleigh, North Carolina, USA.
- M. Brannan, [E. Culf](#), M. Vernooij. “Quantum expanders and property (T) discrete quantum groups.” Invited talk at Workshop on Quantum Graphs, February 2025, Saarbrücken, Germany.
- [E. Culf](#), [A. Mehta](#). “New Approaches to Complexity via Quantum Graphs.” Invited talk at Workshop on Quantum Graphs, February 2025, Saarbrücken, Germany.
- [E. Culf](#), [H. Mousavi](#), [T. Spirig](#). “Approximation algorithms for noncommutative constraint satisfaction problems.” Talk at FOCS 2024, October 2024, Chicago, Illinois, USA.
- [E. Culf](#), [A. Mehta](#). “New Approaches to Complexity via Quantum Graphs.” Talk at TQC 2024, September 2024, Okinawa, Japan.
- [V. Asadi](#), [R. Cleve](#), [E. Culf](#), [A. May](#). “Linear gate bounds against natural functions for position-verification.” Talk at QCrypt 2024, September 2024, Vigo, Spain.
- [E. Culf](#), [A. Mehta](#). “New Approaches to Complexity via Quantum Graphs.” Talk at IWOTA 2024, August 2024, Kent, UK.
- [E. Culf](#), [A. Mehta](#). “New Approaches to Complexity via Quantum Graphs.” Invited talk at Quorum 2024 Meeting, May 2024, Strathmere, Ontario, Canada.
- [E. Culf](#), [H. Mousavi](#), [T. Spirig](#). “Approximation algorithms for noncommutative constraint satisfaction problems.” Talk at IQC-PQCT Workshop, May 2024, Waterloo, Ontario, Canada.
- [E. Culf](#), [H. Mousavi](#), [T. Spirig](#). “Approximation algorithms for noncommutative constraint satisfaction problems.” Poster at QIP 2024, January 2024, Taipei, Taiwan.
- [E. Culf](#), [A. Mehta](#). “New Approaches to Complexity via Quantum Graphs.” Poster at QIP 2024, January 2024, Taipei, Taiwan.
- [E. Culf](#), [A. Mehta](#). “New Approaches to Complexity via Quantum Graphs.” Talk at IQC CS-Math Seminar, November 2023, Waterloo, Ontario, Canada.
- [E. Culf](#), [T. Vidick](#), [V. Albert](#). “Group Coset Monogamy Games and an Application to Receiver-Independent QKD.” Talk at QCrypt 2023, August 2023, College Park, Maryland, USA.

- A. Broadbent, E. Culf. “Uncloneable Primitives with Interaction.” Poster at QCrypt 2023, August 2023, College Park, Maryland, USA.
- E. Culf. “Coset States in Uncloneable Cryptography.” Talk at CMS Summer Meeting, June 2023, Ottawa, Ontario, Canada.
- A. Broadbent, E. Culf. “Uncloneable Cryptographic Primitives with Interaction.” Poster at QIP 2023, February 2023, Ghent, Belgium.
- E. Culf, T. Vidick, V. Albert. “Group Coset Monogamy Games and an Application to Receiver-Independent QKD.” Poster at QIP 2023, February 2023, Ghent, Belgium.
- Anne Broadbent, E. Culf. “Rigidity for Monogamy-of-Entanglement Games.” Talk at ITCS 2023, January 2023, Boston, Massachusetts, USA.
- Anne Broadbent, E. Culf. “Rigidity for Monogamy-of-Entanglement Games.” Talk at CMS Winter Meeting, December 2022, Toronto, Ontario, Canada.
- A. Broadbent, E. Culf. “Coset State Uncloneability to Receiver-Independent QKD.” Poster at Fall INTRIQ meeting, October 2022, Bromont, Québec, Canada.
- Anne Broadbent, E. Culf. “Rigidity for Monogamy-of-Entanglement Games.” Talk at QCrypt 2022, August 2022, virtual.
- E. Culf. “Rigidity for Monogamy-of-Entanglement Games.” Talk at QuICS-IQC seminar, July 2022, Waterloo, Ontario, Canada.
- A. Broadbent, E. Culf. “Rigidity for Monogamy-of-Entanglement Games.” Poster at Spring INTRIQ meeting, May 2022, Bromont, Québec, Canada.
- A. Broadbent, E. Culf. “Rigidity for Monogamy-of-Entanglement Games.” Talk at QIP 2022, March 2022, Pasadena, California, USA.
- A. Coladangelo, E. Culf, J. Liu, Q. Liu, T. Vidick, M. Zhandry. “Hidden Cosets and Applications to Unclonable Cryptography.” Talk at QIP 2022, March 2022, Pasadena, California, USA.
- A. Broadbent, E. Culf. “Rigidity for Monogamy-of-Entanglement Games.” Poster at Quantum Days, February 2022, virtual.